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On using experiments on *remember* as evidence in formal semantics

ABSTRACT

The German memory predicates *noch wissen* ('still know') and *sich erinnern* ('REFL remember') are each used to report two kinds of remembering that are commonly considered clearly distinct in philosophy and cognitive science: experiential remembering of personally experienced events and fact-only remembering of propositions. This methodological paper brings together my existing semantic and experimental work on memory reports involving these predicates. It discusses the questions (i) to which extent experiments on memory predicates can be decisive between formal semantic accounts and (ii) which methodological challenges arise from the multi-disciplinary nature of debates on different kinds of remembering. I argue that to do justice to the usage of memory predicates, individual judgements about other people's cognition must be given a place in the composition. However, any semantic account fulfilling this requirement becomes harder to falsify and the decision between many possible formal accounts is reduced to a modelling choice.

KEYWORDS

memory reports; knowledge; evidence; experiential memory; experimental methodology

1. Introduction

This paper starts with the question whether there is one kind of remembering, linguistically speaking, and later moves on to discuss the methodological challenges of answering this question empirically. The German memory predicates *noch wissen* (‘still know’) and *sich erinnern* (‘_{REFL} remember’) are each used to report two (pre-theoretic, intuitive) kinds of remembering: experiential remembering of personally experienced events (the *wie*-version of (1b) and (2b)) and fact-only remembering of facts about the world, (1a) and (2a).¹

- (1) a. Mia weiß noch {*dass/#wie*} Juist eine Insel ist.
 Mia knows still that/how Juist an island is
 ‘Mia remembers that Juist is an island.’ (German, R&L 2025b: 1a)
- b. Mia weiß noch, {*dass/wie*} Paul im Meer geschwommen ist.
 Mia knows still that/how Paul in-the sea swim is
 ‘Mia remembers Paul swimming in the sea.’ (German, Rosina – Liefke 2025b: 1b)
- (2) a. Mia erinnert sich {*dass/#wie*} Juist eine Insel ist.
 Mia remembers _{REFL} that/how Juist an island is
 ‘Mia remembers that Juist is an island.’ (German)
- b. Mia erinnert sich {*dass/wie*} Paul im Meer geschwommen ist.
 Mia remembers _{REFL} that/how Paul in-the sea swim is
 ‘Mia remembers Paul swimming in the sea.’ (German)

1 The *dass*-version of (1b) and (2b) can be analysed as either, pre-theoretically, see the discussion in Section 4.2.

‘Juist is an island’ in (1a) and (2a) can be replaced with ‘223 is a prime number’ or ‘Napoleon died in 1821’ to make the reader’s clearest case of a non-eventive or not directly experienced proposition. All of these can still be indirectly experienced in the broad sense that there is a learning event, of course. This is central to our (Rosina – Liefke 2025b) semantics presented in Section 2. What matters is that ‘the remembered thing’ in (1a) and (2a) is not *itself* an event. E.g. ‘223 is a prime number’ is arguably not spatio-temporally extended (or its temporal trace is unbounded), and ‘Juist is an island’ only insofar as the island was named Juist at some point. I thank an anonymous reviewer for pushing me to comment on this.

Unless specified otherwise, judgements about the acceptability of sentences in this paper are my own, checked informally with some colleagues. The experimental judgements presented in Section 3 support many of the general intuitions with slightly different sentences.

These two kinds of remembering are commonly considered clearly distinct in philosophy and cognitive science (Bernecker 2010; Cheng – Werning 2016; Devitt et al. 2017 a.o. going back to Tulving 1972): Experiential (‘episodic’) remembering (e.g. of one’s 18th birthday party) involves constructive episodic simulation that comes with brain activity in the left hippocampus and posterior visuospatial regions (Addis et al. 2007) with the result of a detailed, often quasi-visual, phenomenology. Fact-only (‘semantic’) remembering, on the other hand, (e.g. of the fact that singer Cher’s birthday is May 20) is in line with the otherwise outdated metaphor of the human mind as a storage that preserves bits of information in a propositional form.

Given this distinct cognitive and phenomenological status, it is surprising that both common German memory predicates – *noch wissen* (‘still know’) and *sich erinnern* (‘REFL remember’) – report both kinds of remembering, and – philosophically speaking – that we even conceptualize them as two subkinds of the same kind. In Rosina – Liefke (2025b), we stress in particular the need to explain why *noch wissen* (‘still know’) – which appears to be a predicate of knowledge on the morphological surface – can be used to report experiential remembering of personally experienced events (and not e.g. just of the fact that the event in question occurred at all).

Our compositional solution to this puzzle unifies experiential and fact-only uses of *noch wissen* and analyzes the *wie*-complement of experiential uses of *noch wissen* as an informationally rich proposition.² Another important feature of our semantics is that it captures the higher demands for experiential remembering (compared to fact-only ‘remembering that’) through higher standards for *evidence*. (It will become clear in Section 2 how these two features come together.)

In order to check the generalizability of our intuitions in (1) and (2) that led to our semantics in Rosina – Liefke (2025b) and test empirically whether there are natural language markers of experiential remembering, we designed a series of six online studies that are published in Rosina – Liefke (2025a). The results are in line with our initial intuitions that *noch wissen* and *sich erinnern* can report both experiential and fact-only remembering, and that experiential remembering is linguistically reflected in *wie/how*-complements.

While these findings support the need for *some* different treatment of ‘remembering that’ and ‘remembering how’, our semantic claim that both

2 This means that under our analysis the *wie*-complement in (1b) and (2b) is not formally an event anymore under most conceptions of events, see footnote 5.

cases are propositional does *not* in the end capture the philosophical distinction of ‘episodic’ (i.e. experiential) and ‘semantic’ (i.e. fact-only) remembering understood cognitively or phenomenologically. Rather, the phenomenal nature of an instance of remembering in the cognitive sense (fact storage versus constructive/episodic) plays a role as a part of the evidence an agent has for an (informationally rich or poor) proposition.

Furthermore, it remains up for debate how strongly the empirical results (published in Rosina – Liefke 2025a) actually support our formal semantics in Rosina – Liefke (2025b) specifically. What determines (given a stable context) whether someone ‘remembers how’ on our semantics of *noch wissen, wie* (‘still know how’) is whether their cumulated evidence suffices (in the eyes of the speaker) to exclude that things were relevantly different during the event in question from how they actually were. I argue that the importance this gives to the speaker’s judgements of the agent’s cognitive states accurately captures the complex nature of a shared core of memory predicates (henceforth: *remember*), but at the same time necessarily makes it such that our concrete account – like competing ones that also take the speaker-dependency seriously – becomes extremely hard to falsify or support empirically.

In this paper, I aim to bring our compositional semantic account of *noch wissen* (Rosina – Liefke 2025b) closer together with our empirical results (Rosina – Liefke 2025a), asking conceptual, methodological and interdisciplinary questions. Going beyond its linguistic contribution to the semantics of memory reports, I also employ a meta-perspective on linguistics applied to philosophically complex concepts like remembering. Two aspects are central to my investigation:

First, the discussion of two different experimental paradigms with seemingly contradictory results will bring to light the many pragmatic factors that influence the results in Rosina – Liefke (2025a). The difference between the two kinds of experimental paradigms suggests that the way we set up our acceptability judgement experiments successfully tracks semantics rather than pragmatics in the sense of Gricean competition between sentences – but there is no obvious experimental paradigm that can factor out subject-dependency.

Second, I claim that the attitude of remembering in general involves reasoning about different kinds of evidence (cf. Ünal – Papafragou 2020). When speakers use memory reports in natural language, they cannot factor out their beliefs about cognition and even trustworthiness of the agents of memory reports, and we usually do not know how trustworthy our participants find

the agents in our experiments or what their standards for evidence or direct experience are.

This paper is structured as follows: Section 2 gives the necessary background on our (Rosina – Liefke 2025b) semantics of the German memory predicate *noch wissen* ('still know'). Section 3 contains a brief presentation of some of our experiments and findings (Rosina – Liefke 2025a). Section 4 discusses the status of our experiments as evidence for our semantics along two main lines: Section 4.1 discusses the seemingly conflicting results of our two kinds of experimental paradigms. Section 4.2 asks whether the evidential nature of our semantics is supported, and Section 5 concludes and identifies some open issues.

2. Upshot of a composition of *noch wissen*

The question whether there is one kind of remembering can be understood as the question whether there is one lexical-semantic core shared by memory predicates like English *remember*, German *sich erinnern*, and German *noch wissen*, even though they look very different on the morphological surface. For example, *noch wissen* clearly decomposes into *noch* ('still') and *wissen* ('know'), which is the starting point for our semantics in Rosina – Liefke (2025b) that is briefly introduced below. While this 'shared core' hypothesis is no doubt an interesting one, the only strong support for it could come from a broad cross-linguistic investigation that is beyond the scope of my current research. However, the experiments sketched in Section 3 suggest that the compositional semantics of *noch wissen* from Rosina – Liefke (2025b) that is briefly introduced below is at least a viable candidate for such a cross-linguistic *remember*'.

This question is, however, orthogonal to the question whether many of these memory predicates are ambiguous or polysemous along the experiential/fact-only distinction. The composition proposed in Rosina – Liefke (2025b) answers this question negatively for the German memory predicate *noch wissen* ('still know') combining with *dass* ('that') and *wie* ('how') clauses. Instead, the two cases are unified propositionally. The two transparent morphemes of German *noch wissen* ('still know') have been given a lexical semantics in extensive existing literature. Seeing if we get close to the desired truth conditions of (experiential and fact-only) remembering by composing them is a concrete goal in a messy and understudied area. If we succeed in modelling

also the distribution of other memory predicates on the way, this supports our idea that *noch wissen* might be the morphologically transparent version of *remember*.³ (That *remember* and *sich erinnern* behave very similar to *noch wissen* in our rating experiments, sketched in Section 3, at least points in that direction.)

To anticipate, we predict that ‘remembering how’ requires better evidence than ‘remembering that’ because *how/wie*-propositions are more informationally rich (e.g. containing information about details of Paul’s concrete swimming event) and therefore ‘harder to prove’.³ Below, I will explain in a bit more detail how we get to this result. For the full composition, the reader is referred to Rosina – Liefke (2025b).

Before our semantics in Rosina – Liefke (2025b), a polysemous semantics of *remember* (Liefke 2023) was the only suggestion that allowed for *dass*- (‘that’) and *wie*- (‘how’) clauses having the same, propositional, type, while at the same time accounting for the episodic/experiential nature of remembering events from one’s own past. Starting from an observation about German *noch wissen*, we developed a unified propositional account of remembering as a kind of knowledge. *noch wissen* (‘still know’) is very common as a matrix predicate in German when people report past experiences (data from Zöllner et al. 2023), and it is used both with *dass* (‘that’) and *wie* (‘how’) complements. This is exemplified by (1b), the *wie* version of which is repeated as (3).

- (3) Mia weiß noch, wie Paul im Meer geschwommen ist.
 Mia knows still how Paul in-the sea swim is
 ‘Mia remembers Paul swimming in the sea.’ (German)

Our experiments (presented in Section 3) support the intuition that sentences like (3) (as opposed to the *dass*-equivalent) are more acceptable if Paul’s swimming is a part of Mia’s personally experienced past than if Mia was only told about Paul’s swimming. This makes *noch wissen*, *wie* a candidate

3 More precisely, for all *how/that*-pairs of propositions that are identical except for the complementizer, the *how*-proposition is at least as hard to prove as the corresponding *that*-proposition. This falls out as an intuitive consequence of the concept of prove and the entailment of *that*-propositions in their *how*-counterparts, see the explanation of Figure 1 below. (If you prove something, you can also prove whatever is entailed in it in most logics.)

for a morphosyntactic realization of experiential (‘episodic’) remembering as discussed in philosophy.⁴

The high acceptability of (3) reporting experiential remembering is surprising from a semantic point of view given a few standard assumptions. It is often assumed that the object of experiential remembering of personally experienced events is fundamentally different from the object of fact-only remembering, with only the latter possibly a form of knowledge. Non-manner *wie/how*-clauses (Umbach et al. 2022) seem to mark experientiality. So, the first surprise is that something that appears to be a knowledge predicate can embed these *wie*-clauses at all – does that mean that there is experiential knowledge? If knowing is understood as an attitude towards propositions and events are given a non-propositional semantic type, experiential *noch wissen* cannot be analyzed in a morphologically transparent way without running into a type conflict.

Our way to solve this puzzle in Rosina – Liefke (2025b) is to give (pre-theoretic) events a propositional type by analyzing them as informationally rich propositions.⁵ That way, we can give *noch wissen* a unified semantics for fact-only and experiential cases that is close to regular propositional knowledge while at the same time accounting for the informational richness of experienced

4 The same holds for *sich erinnern*, *wie* (‘REFL remember how’), which was not the focus of Rosina – Liefke (2025b).

5 In Rosina – Liefke (2025b), we characterize these non-manner *wie*-propositions as type- $\langle s, t \rangle$ representations of situations/event tokens and distinguish them clearly from manner-*wie* event types (the latter can be paraphrased roughly as “the manner/way in which something was done”). A single lexical element *wie* comes with a choice function over both readings.

In order to be *full* representations of a particular event, however, non-manner *wie*-propositions would have to be maximally specified, which does not only raise ontological issues but also is not in line with actual usage: It is not necessary to represent or have evidence for *all* details of an event in order to remember *wie*. This brings non-manner *wie*-propositions closer to both manner *wie* and informationally poor propositions, since all of these are types, not tokens. Hence, a follow-up version of our semantics has the potential to model a continuum of informational richness and manner-likeness. I leave this project for future work but want to stress the desirability to model non-manner *wie*-complexes as types rather than tokens also empirically: Such an approach captures instances of *I remember how someone entered* (Lucas Champollion, p.c.) and generic memories (*how I used to turn right here*), since there are arguably no event tokens of “someone” entering or habitual event tokens. It also makes it easier to deal with the fact that Ann and Beth can “both remember how Grandma swam” (given that they have different experiences and maybe witnessed different temporal parts of the swimming) without having to make too many assumptions about event boundaries and cumulativity.

events. In order to explain that *noch wissen, wie p* entails *noch wissen, dass p* but not vice versa – ‘remembering how’ is more demanding than ‘remembering that’ – we give *wissen* (‘know’) an evidence-based lexical entry and note that having evidence for an informationally rich proposition is much more demanding than for an informationally poor one like ‘Juist is an island’.⁶

The main idea of our semantics is that even experiential remembering *wie* (‘how’) is retained knowledge of a proposition, just of an informationally rich proposition (‘what was the case when *p*’).⁷ In [*noch* [*wissen* [*wie p*]]], *noch* (‘still’) contributes a certain temporal requirement – the same as in ‘it’s still raining’ – *wissen* knowledge, and *wie* turns an informationally poor proposition into an informationally rich proposition.

Our unification of *dass* (‘that’) and eventive *wie* (‘how’) clauses builds on existing literature. Liefke’s (2023) and Umbach et al.’s (2022) semantics of non-manner *how* provides the tools for capturing both cases as propositions, and the way our semantics links kinds of evidence with *wie* (‘how’) shows clear parallels to perception reports (Davis – Landau 2021) and the general debate on evidentiality in cognition reports (Ünal – Papafragou 2020). Due to its unification of experiential and fact-only remembering, our semantics makes sense of the observation that the two cases are expressed by the same

6 As a consequence, all of our following claims about the interplay of evidence and informational richness are not only about *noch wissen* (and potentially about *remember*’, if our generalization holds), but also about bare *wissen* (‘know’). While we have not explored the consequences of this general evidential semantics of knowledge much further, we think that the corresponding prediction of informationally rich *wie/how*-complements without *noch* (‘still’) is born out in the case of evidentially supported present-directed imagining. Consider (i) in the context of having watched the neighbors snap at each other and storm into their house (= Rosina – Liefke 2025b: (14b)):

(i) Ich weiß (genau/schon/ja/#noch), wie die jetzt (wieder) aufeinander losgehen.
I know (exactly/PART/PART/still) how these now (again) at-each-other attack
‘I know (exactly) how these two are fighting (again) right now.’ (German, Rosina – Liefke 2025b: 14b)

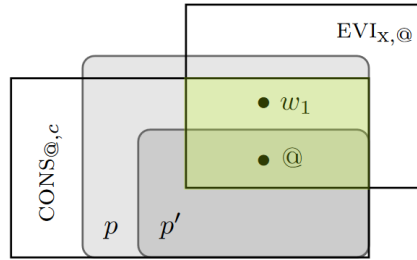
7 ... where which aspects of the salient ‘time when *p*’ are relevant is determined depending on the context. This makes much weaker predictions than a semantics that demands recall of certain properties of the *event* in question (under most notions of events). I think that this consequence is welcome to capture collective memories. We predict the following sequence to be acceptable: *Paul and Alex only met last year, but they both remember how Neil Armstrong landed on the moon. Paul watched it on TV with his parents, but Alex mainly remembers their sister’s excitement on that day.* I thank Nina Haslinger (p.c.) for discussing these cases with me and an anonymous reviewer for pointing me to their relevance here.

lexical item in English *remember* and the two German memory predicates discussed in this paper. It is independently supported by predicting the intuition that *genau* in *noch genau wissen*, *wie* is best translated as ‘vividly’, and in *noch genau wissen*, *dass* as ‘for sure’.⁸

An intuitive way to grasp our semantics is to focus on four factors that contribute to someone *not* remembering *dass p* or *how p*. Consider our evidence-based entry for *wissen* (‘know’) in (4), formalizing Lewis (1996), using the common presupposition/assertion distinction for ‘know’ from Kiparsky – Kiparsky (1970), and the model in Figure 1:

- (4) $[[\text{wissen}]]^{\text{@},c} = \lambda p: p_{\text{@}}. \lambda x. \text{CONS}_{\text{@},c} \cap \text{EVI}_{x,\text{@}} \subseteq p$
 ‘(*p* holds in the actual world and) *p* holds in all considered worlds (CONS) that are compatible with the agent *x*’s evidence (EVI).’

Fig. 1. A model of *x* knowing *p* while not knowing *p*’



EVI is the set of worlds that is compatible with agent *x*’s cumulated evidence. It is not equivalent with *x*’s doxastic alternatives, because it is possible to have access to a piece of evidence (i.e. what the speaker considers evidence wrt. the topic of *p*) without using it for restricting one’s belief set.⁹ Whenever an agent *x* encounters a new piece of evidence (that is a new piece of information that the speaker considers causally relevant wrt. accepting or rejecting *p*), the set $\text{EVI}_{x,\text{@}}$ shrinks.

8 ... using Wurm’s (2020) semantics of *genau* for both cases. See Rosina – Liefke (2025b) for the argument.

9 This accounts for instances of ‘come on, you know it!’. Intuitively, we can use this utterance in an accusatory way while being fully aware that the interlocutor does not believe whatever ‘it’ refers to. So, knowledge without belief is possible under our account, and knowledge is not transparent. So, one can know something while not knowing that they know it – for example because of suppressing their knowledge as a defense mechanism.

When I talk of ‘having good evidence’ or ‘having enough evidence’ in the rest of the paper, what I mean is always ‘having evidence such that EVI is small enough for the given case’, which does not distinguish between these two aspects: I take it that a speaker agrees that an agent has enough pieces of evidence iff they agree that the agent has good cumulated evidence. To illustrate this idea of evidence, consider the case of Anna and Bea, who agree that Paul was swimming in the sea, but have different beliefs about this swimming event. Anna eye-witnessed Paul’s swimming ten years ago while Bea read a detailed and reliable report of Paul’s swimming a week ago. Who of the two has better evidence for ‘how Paul was swimming in the sea’ (on a non-manner reading)? According to my intuition, this is Anna – because even if her memory is fading, she would probably still be a better painter of a picture of the swimming event – maybe lacking many factual details like the time of the day or the color of Paul’s pants, but able to recollect the angle Paul’s body had in the water, roughly how far from the coast this was, Paul’s hairstyle at the time, a particular way of smiling he had back then etc. These details are very hard to describe in a report or usually not considered important in a report of an event. Of course, there is a theoretical flipping point to this: If Anna turns out to have been high on hallucinogenic drugs back then and the report Bea reads is 2000 pages long, Bea might be the one who can utter *I remember (from the book) how Paul was swimming in the sea* at least better than Anna can utter *I remember (from my drug-influenced recall) how Paul was swimming in the sea*, if not perfectly. While it is hard to have clear intuitions about such artificial cases, the reader can check whether their linguistic and evidence-judging intuitions align: If one thinks that Bea has better evidence for the (perceptual etc.) details of the swimming event, then one should – according to our semantics – also have the intuition that she is the better speaker of the *how*-sentence. Evidence must always be understood as evidence *for something*. Hence, while we may consider a detailed report to be very good evidence out of context, most reports are not very good evidence when it comes to perceptual details like lightning. Our semantics is more about “what the evidence is for”, whether it is suitable evidence for an informationally rich proposition, than about strength in an absolute sense. This is captured by the interaction of EVI with other elements in our composition.

CONS is the set of possible worlds that have to be considered as world-candidates pre-evidence in a given context. It is inspired by Lewis (1996)’s observation that the context-dependency of the standards for knowledge can be

captured by modulating which possible worlds can be legitimately ignored. E.g. in an everyday situation, an agent does not usually have to look for evidence to exclude that they are a brain in a vat to make acceptable knowledge statements about what they directly perceive. But in an epistemological discussion, they might have to do exactly that. The epistemological context would come with such an unrestricted, large set CONS – every remote possibility has to be considered – that EVI alone has to do the job of restriction (since what counts is their intersection, see Figure 1).

For a (*noch*) *wissen* report to be true, our semantics demands that the intersection of EVI and CONS – the green area in Figure 1 – is a subset of p . Paraphrasing informally, this means that for all contextually relevant, considered not- p worlds, x 's cumulated evidence has to be good enough to exclude them as a candidate for the actual world. In the model in Figure 1, the agent x knows p but does not know p' , because the green area is a subset of p , but not of p' . We identify *dass* p with p in Figure 1, and *wie* p with p' in Figure 1, so the agent knows *dass* p , but not *wie* p . We achieve this by treating *wie* as an informational richness operator (Liefke 2023; inspired by Umbach et al. 2022) sending an informationally poor proposition '(that) Grandma was swimming in the sea' to an informationally rich proposition p' corresponding roughly to 'how things were when Grandma was swimming in the sea'. This is a subset of p (which subset exactly is – again – context-dependent and can be different for a court case than for gossiping). An interplay of four different factors can prevent remembering, three of which are due to *wissen* and its complements:

- (I) EVI can be “too large”, i.e. the agent does not have enough evidence.
- (II) CONS can be “too large”, which is often the case e.g. in epistemological or court contexts, and in the case of modification by ‘exactly’, see our discussion in Rosina – Liefke (2025b) applying Wurm (2020).
- (III) p/p' can be “too small”. The more informationally rich a proposition, the harder it is to be known. Since *wie* p is always a subset of p , it is harder to ‘know how p ’ than it is to ‘know that p ’ for any fixed p . For example, agent x in Figure 1 cannot exclude worlds like w_1 where e.g. Grandma did swim in the sea, but under circumstances that conflict with “the way things were when she actually did”. In order to exclude these extra worlds, the agent would have to compensate for the small size of p' by collecting extra evidence and thereby shrinking EVI .
- (IV) Finally, even when knowledge is given, it can fail to be retained in the right way, as specified in our composition with a standard, temporal

entry for *noch* ('still'; simplified from Beck 2019; Löbner 1990; Krifka 2000): *x weiß noch, wie p* presupposes that *x* has had evidence for the informationally rich *p'* in a preceding time interval.¹⁰

To conclude this section, let us take stock of what we have and have not achieved with our semantics of *noch wissen, wie p*. A semantics of experiential memory reports should make sense of the inferences from (3) to (5a)–(5c).

- | | |
|---|--|
| (5) a. Mia remembers what it was like when Paul was swimming in the sea: (he was wearing blue shorts, his body was being tossed about by the waves ...) | Informational Richness,
(D'Ambrosio – Stoljar 2023 a.o.) |
| b. Mia has (perceptually and/or emotionally) experienced Paul swimming in the sea. | Experientiality,
(De Brigard 2023 a.o.) |
| c. Mia remembers that Paul was swimming in the sea. | that-Entailment |

We account for the inference to (5a) by analyzing *wie* as an informational richness operator in Liefke's (2023) sense, and the inference to (5c) follows by (III) above. We do not, however, account in our semantics for the common claim that previous direct experience of the remembered event is necessary for experiential remembering (Stephenson 2010 a.o.). Our story for the common (not necessary, on our account) inference to (5b) is that it is a consequence of world knowledge. What follows semantically is an *EVI* that is smaller than in the *dass/that* case (given that *CONS* is fixed contextually), i.e. high standards for evidence. If we assume that in the actual world, direct experience is usually considered the best kind of evidence (at least for informationally rich propositions), then it follows for many speakers that in order to 'remember how' one has to have directly experienced the event in question. Consequently, (5b) is not reflected in the semantics of *noch wissen* directly but is instead a result of individual standards and beliefs. If a particular speaker does accept hearsay as evidence for an informationally rich proposition, they *will* grant the agent that they 'still know how'. I will pick up this discussion again in Section 4.2. For the upcoming experimental Section

10 The negative counterpart of *noch, nicht mehr* ('not anymore') shares the presupposition of *noch* and asserts the negation of the assertive content of *noch*. This correctly predicts that *nicht mehr wissen* ('not knowing anymore') amounts to having forgotten.

3, this means that I will call German *wie* and other candidates (potential) ‘e-markers’, where ‘e’ is underspecified between ‘experience’ and ‘evidence’.

3. Experimenting with experiments

Our experimental data (Rosina – Liefke 2025a) supports the status of *wie* as an e-marker as well as parallels between German *noch wissen* and *sich erinnern* and contains pointers to a broader phenomenon of e-markers in memory reports. The latter is because English *remember* plus gerund – *remember Paul swimming* as in the translations of (1b) and (2b) – behaves similarly. The experimental support of some of the intuitions mentioned so far as well as pointers to a possible generalization to a shared core of memory predicates are the topic of Section 3.1.

For the case of *dass/that* complements in situations where the agent only has been told (by a reliable source) about the event (so concerning the question whether indirect experiencers have fact memory), our semantics (Rosina – Liefke 2025b) predicts acceptability of *x remembers that p*. – However, this prediction only carries over to actual experiments as far as pragmatic competition between ‘remembering how’ and ‘remembering that’ and with other attitude reports can be kept out of the judgements. In fact, as presented in this Section 3, the results depend heavily on the experimental paradigm. Section 3.2 introduces two complementary experiments that are relevant for these considerations in short. (In the later Section 4.1, I suggest that pragmatic effects interact in different ways depending on the experimental paradigm.)¹¹

3.1 Attesting e-markers

With the help of Kristina Liefke and our student assistants, I designed an experimental paradigm that is aimed at detecting e-markers in memory reports. The detailed experimental set-up appeared in Rosina – Liefke (2025a), can be experienced directly with our mock version of the English version of the experiment available at tinyurl.com/e-markers, and is prere-

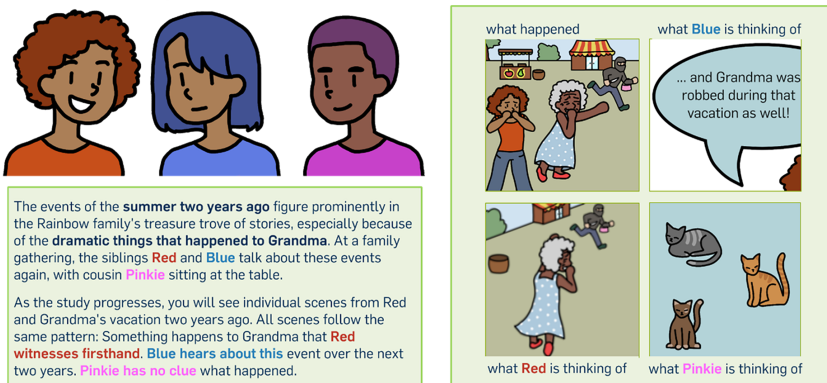
11 This experimental Section 3 has profited greatly from the detailed comments of an anonymous reviewer.

gistered in the case of the German *sich erinnern* rating experiment (Rosina – Liefke 2024a).

We ran two German experiments of the general form described in this section (with the predicates *noch wissen* and *sich erinnern*, both with *wie* complements as hypothesized e-markers) as well as an English version with the same experimental setup (with the predicate *remember* and gerund complement clauses – like in *remember Grandma swimming* – as hypothesized e-markers). I will only describe the German rating experiment on *sich erinnern*, *wie* (‘REFL remember how’) in a comprehensive way in this paper. The other two versions are described in Rosina – Liefke (2025a). This means that the claim that the three predicates (*noch wissen*, *remember* and *sich erinnern*) behave similarly is only supported through graphs (Figures 3–5 below) here and not much discussed. Instead, I will present our experimental paradigm along the lines of the question whether our semantics of *noch wissen*, *wie* (as summarized in the previous Section 2) as retained evidential knowledge of an informationally rich proposition is a suitable candidate (awaiting future cross-linguistic investigation) for a generalized core of memory predicates, *remember*?. This is exemplified here by a possible extension to the other German memory predicate, *sich erinnern*.

Our rating experiment introduces participants to the background story of a family gathering and a set of teenage cousins who represent different types of experience/evidence with respect to different past events of something bad happening to their grandma. For the present purpose, the most important distinction to bear in mind is between the teenagers RED and BLUE in Figure 2.

Fig. 2. Composition of screenshots and text from the English version of the experiment



Participants are instructed to judge the acceptability of highlighted sentences against a background of ‘what happened’ and ‘what [one of the characters] is thinking of [two years after the event]’ (all combined in one vignette). For the robbery scene in Figure 2, this means that they see the ‘what happened’-picture plus one of the teenagers’ mnemonic perspectives. Red is always the direct experiencer, Blue the indirect experiencer, and Pinkie the control character who is thinking of cats and other unrelated things. Below this combination of two pictures and a reminder text, one of the following sentences appears in the target vignettes (for the robbery scene, and sentences with the corresponding complements for the other scenes):

- (6) a. Red sagt: Ich erinnere mich, dass Oma überfallen wurde.
 b. Blue sagt: Ich erinnere mich, dass Oma überfallen wurde.
 c. Red sagt: Ich erinnere mich, wie Oma überfallen wurde.
 d. Blue sagt: Ich erinnere mich, wie Oma überfallen wurde.
- {Red/ says I remind REFL {that/ Granny robbed was
 Blue} how}
- {‘Red/Blue} says: I remember {that/how} Granny was robbed.’ (German)

For each vignette (i.e. a combination of two pictures and a target or filler sentence), Participants were asked to provide ratings on ‘The sentence marked in green, uttered by [SPEAKER], describes the situation ...’ – on a scale from 1 (‘not correctly at all’) to 7 (‘absolutely correctly’).¹²

As deducible from (6), both the character uttering a sentence (RED, BLUE) and the complementizer (DASS, WIE) are manipulated variables in the main experiment, leading to four conditions, such that for each scene (in this case, of Grandma being robbed) we get a set of four target vignettes. Alongside the robbery scene, we used three more target scenes: Grandma swimming and almost drowning, falling out of a canoe, and burning a cake. Hence, we tested 16 target vignettes (augmented with 16 filler vignettes).¹³

12 In German: ‘Der grün hinterlegte Satz, von [SPEAKER] gesagt, beschreibt die Situation ...’ – on a scale from 1 (‘gar nicht richtig’) to 7 (‘völlig richtig’).

13 False fillers attribute Pinkie – who has no evidence or experience wrt. the event whatsoever – remembering of the events involving Grandma. True fillers attribute Red perception of these events or are unrelated true statements about Pinkie like that he is thinking about cats when this is depicted by the Pinkie-picture, see Figure 2.

Target sentences and fillers were assigned blocks. Both the order of these blocks as well as the order within each block was randomized. This arrangement prevented unhelpful

We recruited via Prolific, excluded six participants from analysis based on a filler performance of $\leq 75\%$ and analyzed data from 54 German mono-lingually raised native speakers. All of them provided judgements on all 32 vignettes (so we did not use a latin square design) in order to enable us to make all pointwise comparisons per participant (in case of big individual differences).¹⁴

The results support both of our hypotheses in (7) and the effects we found are highly statistically significant. We consider this evidence for *wie* as an e-marker in German *sich erinnern* reports. This is what is predicted by an extension of our semantics for *noch wissen* reports (sketched in Section 2) to these cases, because the informationally rich *wie*-complement makes it such that the speaker needs evidence not only for the fact that Grandma was robbed, but for details of the event – evidence that only RED can have given our background story.¹⁵

clustering of minimal pairs while also ensuring that the order varies between participants (to control for effects of order to some degree).

- 14 No serious issues with concentration were reported in our post-survey questions.
- 15 All of our analyses use Cumulative Link Mixed Effect Models fitted with the Laplace approximation. For motivation of this choice, see Liddell – Kruschke (2018). In the case of pointwise comparisons – which is closest to the formulation of our hypotheses in (7) – we filtered the dataset for *WIE* (7a) or *BLUE* (7b) and contrasted the remaining two ‘markerspeaker’ conditions (hence, ignoring their 2x2-nature) for each case with one another using the model $\text{responseO} \sim \text{markerspeaker} + (1 \mid \text{Participant}) + (1 \mid \text{scene})$ and ± 0.5 sum contrasts. Our focus on this pointwise comparison is due to a decision for a preregistration strategy that was both cautious and as precise as possible wrt. expectations from our semantics: According to our design and background story, Red clearly has *better* evidence for the *how/wie*-propositions than Blue does (hence, our semantics predicts HYP. I, 7a). Furthermore, Blue does not have enough evidential support for the *how/wie*-propositions, but her reliable hearsay evidence should suffice as evidence for the fact that the event happened at all (hence, our semantics predicts HYP. II, (7b)). While, viewed purely semantically, we also predict RED+DASS to score similarly high as RED+WIE and BLUE+DASS, we were not sure whether this truth-conditional expectation would be reflected in the experiment due to pragmatic factors: If the scale task would give rise to competition between the two possible sentences for Red, RED+DASS might score lower than BLUE+DASS. For this reason, we preregistered no hypothesis building on RED+DASS in any way – so no pointwise comparisons involving RED+DASS and also no hypothesis on interactions between the two factors.

A reviewer notes correctly that the pointwise hypotheses could also have been tested with two single-factor designs. We did, however, announce testing with models for interactions in addition to the pointwise comparisons in our preregistration. While this is of a somewhat explorative nature, it was important to us to be able to perform these additional analyses (given that our initial hypotheses would be confirmed) to also find out whether there was an interaction between the factors. Accordingly, we made use of the



- (7) a. ✓ HYP. I: Higher ratings for RED+WIE than for BLUE+WIE ***
b. ✓ HYP. II: Higher ratings for BLUE+DASS than for BLUE+WIE ***

In two other experiments (see Rosina – Liefke 2025a), we backed up the equivalents of Hypotheses I and II in (7) for ‘noch wissen’ reports with ‘wie’/‘dass’ complements and for English ‘remember’ reports with gerund (‘remember Grandma swimming’) versus ‘that’+past simple complements. Figures 3–5 visualize the strikingly similar results of all three experiments.

Fig. 3. ‘sich erinnern’, n=54

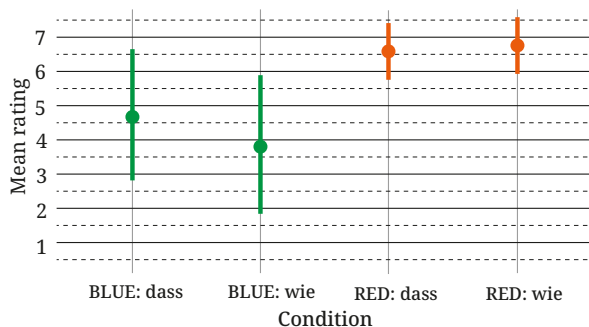
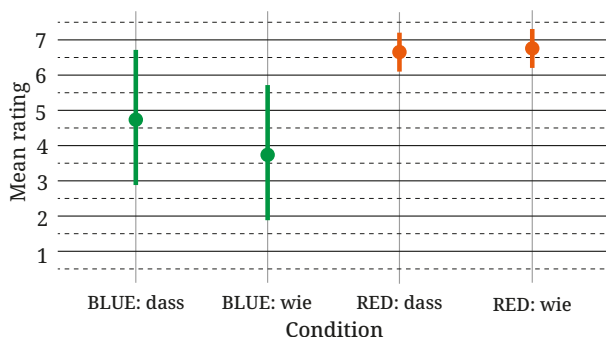
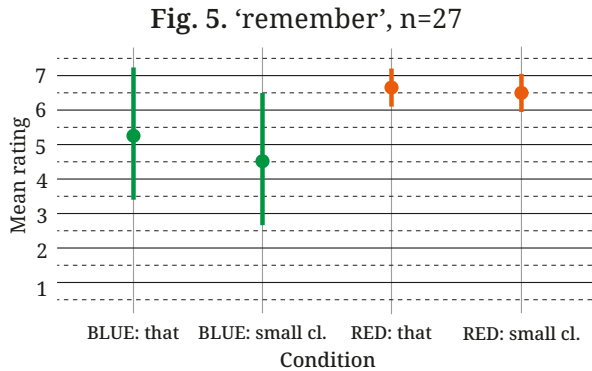


Fig. 4. ‘noch wissen’, n=37



2x2 design by also testing for main effects of both factors and an interaction of SPEAKER and MARKER with ‘responseO ~ speaker * marker + (1 | Participant) + (1 | scene)’, compared with ‘responseO ~ marker + marker:speaker + (1 | Participant) + (1 | scene)’ and ‘responseO ~ speaker + speaker:marker + (1 | Participant) + (1 | scene)’ (where ‘marker’ is the sum-coded MARKER condition and ‘responseO’ the ordinal rating). We found all of these effects to be significant.

Significance codes: 0*** 0.001**0.01 * 0.05 ‘.0.1



Note that our experiments so far provide no reason to characterize e-markers as clear evidentiality markers – as predicted by our (Rosina – Liefke 2025b) semantics, only considering truth-conditions – *or* markers of direct experience as predicted by Stephenson (2010), since RED both has good evidence for the informationally rich proposition ‘how Grandma was robbed’ *and* has directly experienced the corresponding event. If the phenomenon of e-marking could be shown more broadly to range over different memory predicates, complements, and even languages, this would constitute further support that cognitive concepts of evidentiality “precede” its natural language realizations (Ünal – Papafragou 2020). As the empirical picture is now, we need at least very similar use conditions for all of the cases we tested. So far, an extension of our semantics for *noch wissen* is a good candidate for *remember*, since it also models our findings wrt. *sich erinnern* reports.¹⁶

The only thing that remains mysterious at this point are the ratings for BLUE+DASS (for both *noch wissen* and *sich erinnern*, and for BLUE+THAT in the case of *remember*). These are higher than the ratings for BLUE+WIE, which is everything we hypothesized about them (7b). They are, however, lower than

16 An anonymous reviewer points me to the observation that both Mandarin and Classical Latin seem to have *remember*-complements such that the same sentence is a natural translation of both English *that*- and *how*-complements, and that these languages seem to lack e-markers. If this should turn out to be a stable typological property of some languages synchronically, one way of interpreting the ‘remember’-complement is giving it an underspecified semantics (like possibly English DPs as in “remember my birthday”). Speculatively, another route could be to claim phonologically null *how* and *that* in these cases. In either case, these languages do not constitute counterexamples to our semantics but rather would not help decide between our semantics and other proposals at all. (Similar to how languages that don’t mark plural at all don’t tell us much about verbal agreement wrt. number.) Rather, I take our semantics to be supported iff a given e-marker language behaves like German and English wrt. the Blue/Red-judgements.

is explainable by any suggested truth-conditions of memory reports (including ours) alone. Blue does have reliable indirect evidence for the event, so the fact that it happened at all should be evidentially supported, plus direct experience is not required for ‘remembering that’ by Stephenson’s (2010) account either. Our results are interpretable in at least two ways relating to this finding:

Reasoning mainly with the difference between the speakers in mind, one could speculate that some participants do not grant the character with reliable indirect evidence, BLUE, any remembering related to Grandma’s mishaps at all, and consequently not even fact-only ‘remembering that’, while RED is just attributed “good memory” in general with respect to the topics depicted. The effect of the complementizer for Blue (predicted by our semantics) would have to be built on top of this general preference for Red.

Alternatively, one can – in an environment of a general interaction – explain the speaker main effect (see fn. 15) as carried by the low ratings for BLUE+DASS as an independent phenomenon and the high ratings for RED+DASS as an independent phenomenon (the former one unexpected by everyone, the latter one predicted by our semantics given that the experimental set-up tracks truth-conditions and rules out pragmatic competition between RED+DASS and RED+WIE).

I will come back to this phenomenon in Section 4.1.

3.2 Experiments on pragmatic effects

Before settling on the experimental set-up of our rating experiments as described in Section 3.1 for *sich erinnern*-reports, we tested the rating paradigm against a forced choice paradigm in two studies on *noch wissen*-reports. The forced choice study is reported as one of the six experiments in Rosina – Liefke (2025a). We decided to go with the rating paradigm for the broader experiments because it seemed to track truth conditions better (see the discussion in Section 4.1), which is what we were mainly interested in at that time. However, I take the forced choice paradigm to be informative about the semantics-pragmatics interface of memory reports, which is why I discuss it below. We did not do an inferential analysis of the forced choice experiment, nor was it preregistered (both as opposed to the rating experiment on *sich erinnern* presented in Section 3.1 above and to the ‘intermediate evidence’-rating experiment described later in this Section 3.2). Hence,

the findings of the forced choice experiment should be taken with a grain of salt and optimally followed up on with future experiments.

The speaker-identification task of the forced choice experiment was inspired by Davis – Landau’s (2021) experiment on perception reports and makes participants choose “Wer sagt [sentence]?” (‘Who says...?’) – RED, BLUE OR PINKIE – for each of the sentences presented against a given context. The background story is the same as described above for the rating experiments, and the forced choice study uses two of the target scenes (robbery and swimming) of the rating experiments. The sentences are the same as in the *noch wissen* rating experiment and so identical to the green highlighted sentences in (6) for the robbery scene apart from the memory predicate (which was *noch wissen* in this case). This leads to a single-factor design with SPEAKER as the dependent variable and MARKER (i.e. complementizer for the German experiments) the only manipulated variable.

As a participant, one had to judge less vignettes in this study (8 in total, only 4 target vignettes, the same for all participants), but the individual vignettes were more to process.¹⁷ Participants saw all four pictures to the right of Figure 2 (so ‘what happened’ and what all of the teenagers are thinking of) at once. Below, they were presented with one target or filler sentence.¹⁸ We forced participants to choose exactly one speaker of the given sentence, instructing them to choose the one for who it is most likely that they said the sentence, if they think that more than one or none of them could have said it. The results (from 29 German native speakers) support both hypotheses in (8) descriptively:

- (8) a. ✓ HYP. I: RED is selected more often in the WIE condition (84%)
- b. ✓ HYP. II: BLUE is selected more often in the DASS condition (64%)

The support of Hypothesis I (8a) is directly predicted by our semantics of *noch wissen*, since the informationally rich *wie*-complement is expected to come with a higher threshold for cumulated evidence – which is only met by Red. Two more things are noteworthy about these results. First, the support of Hypothesis II is only indirectly related to our semantics: We predict

17 Like in the rating experiments, we received good ratings on the clearness of the task and our instructions.

18 The filler sentences in this case were having seen (Red), heard (Blue) or having no clue (Pinkie) about the event. Additionally, we had decided to exclude participants who would choose Pinkie for any of the target vignettes – which no one did.

that – truth-conditionally – both Red and Blue should meet the standards of ‘still knowing / remembering that’, so without any pragmatic interference, we should have gotten randomly chosen answers. However, we already suspected informally (again, this experiment was not pre-registered) that the forced choice task would give rise to competition, such that Blue ends up being the better speaker of the *dass*-sentences. Note that the difference of BLUE>RED for the *DASS* condition is weaker than the difference of RED>BLUE for the *WIE* condition. If *dass* was an indirectness marker, truth-conditionally speaking, we would have expected a similarly high preference for BLUE in the *DASS* condition.

Second, while the support of Hypothesis II is expected taking the forced choice experiment by itself, it is not relative to the results of our rating experiments. The preference for BLUE>RED in the *DASS* condition clashes with significantly higher ratings for RED+*DASS* than for BLUE+*DASS* in all rating experiments (also ones that are not presented in this paper). If we present participants with the German equivalent of the sentence “I remember that Grandma was swimming in the sea”, they choose BLUE as a speaker. However, if we ask them about this very sentence uttered by BLUE, they rate it lower than when RED utters the same sentence. I will come back to this puzzle in Section 4.1.

To conclude the series of experiments described in this paper, I briefly present a variant of the *sich erinnern* rating experiment presented in Section 3.1 that targets the gradability of the concept tracked by e-markers (that is, most prominently, of kinds or quality of experience or evidence). This experiment makes use of (again the same background story and) a subset of the target scenes of the experiment presented in section 3.1, that are also presented in the same form to the participants.¹⁹ The memory predicate tested is *sich erinnern*, and the green highlighted sentences are the exact same as in (6) – only also uttered by a new character, Goldie, in addition to Red and Blue (and in some fillers Pinkie), as explained below. This experiment was preregistered (Rosina – Liefke 2024b) and has also appeared in Rosina – Liefke (2025a). We collected answers from 40 participants and excluded 4 of them based on filler performance of $\leq 75\%$. Every participant answered all of the 24 vignettes (12 target, 12 filler). Adding GOLDIE as another level of the speaker variable (in addition to the complementizer variable with *WIE*

19 I.e. only the swimming and the cake burning scene; the filler scenes were also the same as in the original *sich erinnern* rating experiment.

and DASS) led to 6 conditions for the target vignettes (so two vignettes per condition).

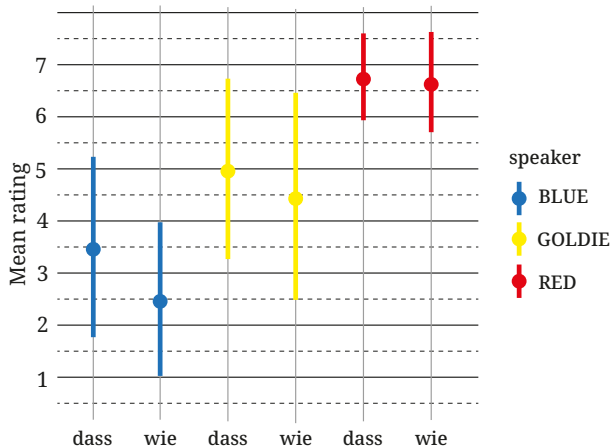
GOLDIE was introduced as another cousin who always missed the things that happened to Grandma by a few minutes but saw the immediate result (like a broken leg). The idea behind GOLDIE is that she has ‘intermediate evidence’, but no direct experience, and so ratings between RED’s and BLUE’s for GOLDIE in the WIE condition would point to the gradability of the concept licensing e-marking we predicted based on our semantic account together with some additional observations and assumptions: If a direct experience requirement was a part of the semantics (like in Stephenson 2010), Goldie should pattern with either Red or Blue, depending on whether she meets the directness requirement (at least for a given individual participant-scene combination). Our (Rosina – Liefke 2025b) account, however, locates experientiality at the level of pragmatics and claims that ‘eventive *wie*’ is only a marker of informational richness that when embedded under predicates of knowledge and remembering, marks good enough cumulated evidence (regardless of the kind of evidence). We reasoned that ‘good enough cumulated evidence’ together with the speaker- and context-dependency of our account would provide more flexibility to deal with gradience. (For complications with this reasoning, see the discussion in Section 4.2.) This is exactly what we found. There were significant main effects of RED>GOLDIE>BLUE, and the DASS/WIE contrast shrinks ‘in the RED direction’, see Figure 6.²⁰ The results support 5 of our 6 hypotheses as we formulated them in the preregistration, see (9). (We just expected Goldie to be a bit closer to Blue in terms of the DASS>WIE effect.) Summarizing their essence informally, we hypothesized a replication of the previous results plus that Goldie would be between Red and Blue with respect to both hypotheses from the original experiment, because the concept licensing e-marking would be gradable.²¹

20 As in the study presented in 3.1, we found all the corresponding interactions using Cumulative Link Mixed Effect Models, but also the pointwise differences when the 3x2 conditions were treated as 6 independent ones. The pointwise comparisons were done in the same way as in the 2x2 studies, see fn. 15. For Hyp. III (9f), we compared the $\Pr(>|z|)$ results of the pointwise models for (9d) and (9e). In addition, we exploratively sum-coded contrasts to compare Blue to Goldie and Goldie to Red and found main effects and interactions.

21 This is also the reason why we did not treat the three character levels as ordinal, because that is in a sense what we were trying to show.

- (9) a. ✓ HYP. IA: Higher ratings for RED+WIE than for BLUE+WIE ***
 b. ✓ HYP. IB: Higher ratings for RED+WIE than for GOLDIE+WIE ***
 c. ✓ HYP. IC: Higher ratings for GOLDIE+WIE than for BLUE+WIE ***
 d. ✓ HYP. IIA: Higher ratings for BLUE+DASS than for BLUE+WIE ***
 e. X HYP. IIB: Higher ratings for GOLDIE+DASS than for GOLDIE+WIE $\Pr(> |z|) = 0.114$
 f. ✓ HYP. III: The effect confirming IIA is bigger than the effect confirming IIB

Fig. 6. Ratings by condition, ‘intermediate evidence’-study



4. Pragmatics as a gateway for epistemology

4.1 A competition of competitions?

Taken together, the experimental results from Section 3 support the existence of e-markers in the complements of predicates of remembering. Given the minimal pairs that we tested, German *wie* (‘how’) is such a marker in opposition to *dass* (‘that’).²² The results of our rating experiments show that *dass/that* is also acceptable for a direct experiencer, and the results of the forced choice experiment suggest that certain experimental tasks give rise

²² For support that gerund *-ing* small clauses are e-markers in contrast to *that* clauses in simple past tense, see Rosina – Liefke (2025a).

to a competition that makes it more likely that a given *dass* memory report was uttered by an indirect experiencer than that it was uttered by a direct experiencer. Hence, the rest of the discussion takes for granted that *dass* is semantically licenced for RED and BLUE, *wie* only for RED. Section 4.2 below addresses the question what makes RED the best candidate (and GOLDIE an ok candidate) for experiential remembering. This Section 4.1 discusses which pragmatic competitions seem to play into our experiments.²³

Speculatively, the relatively low ratings of BLUE+DASS in the rating experiments could be a result of competition with other sentences that BLUE could have said instead. Both (10b) and (10c) are morphosyntactically simpler than (10a),²⁴ so if they convey the same in the context of our study, we expect them to lower the acceptability of (10a) due to a pressure to minimize complexity.²⁵

- (10) a. Ich weiß noch, dass Oma im Meer geschwommen ist.
 ‘I {remember / still know} that Grandma was swimming in the sea.’
 b. Ich weiß, dass Oma im Meer geschwommen ist.
 ‘I know that Grandma was swimming in the sea.’
 c. Oma ist im Meer geschwommen.
 ‘Grandma was swimming in the sea.’

23 Both sections focus on German *wie* and *dass*, but our English experiment suggests that we are going to need a story that is similar in effect for English *gerund-ing* small clauses.

24 For (10b), this is reflected in our semantics for *noch* modifying *wissen* in *noch wissen*. If an extension to *noch wissen* = *remember* holds (in stative cases as opposed to eventive ‘coming to mind’ uses of *remember*), this straightforwardly explains why the same competition appears for mono-morphemic memory verbs.

25 The competing sentences might well be different for different memory predicates – even if their core semantic effect should turn out to be equivalent – due to their morphological make-up. E.g., *noch wissen* (‘still know’) might allow for a more direct contrast with *nicht mehr wissen* than *sich erinnern* (‘REFL remember’) does, and only *sich erinnern* allows for the common ability/dispositional construction *sich erinnern können* (being able to remember; thanks to an anonymous reviewer for pointing out the relevance), while *können* (‘can’) in *noch wissen können* only allows for an epistemic reading. I suspect that this is due to the fact that only *sich erinnern* but not *noch wissen* allows for eventive/actualization readings where the rememberer has to be currently in the process of recalling. Only this reading can be turned dispositional, the evidential one I am discussing in this paper cannot. What is important for the argument here to apply beyond *noch wissen* (and explain our experimental data) is that other memory predicates, too, must enter competition with (at least) knowledge predications (10b) and unembedded assertions like (10c).

Importantly, the competition in (10) is not active for RED (RED+DASS is rated as high as RED+WIE, see Figures 3–6). This means it must be something about BLUE, the indirect experiencer, that makes it relevant for them (at least in the scenes in our experiments) that they stress the quality (‘know’) and retention (‘still’) of their evidence.

Why would (10a)–(10c) convey the same information in the context of our rating experiments, but more so for Blue than for Red? According to our semantics of *noch wissen*, *dass p* (‘still know that *p*’), the truth of *p* in the actual world is presupposed, and having good evidence for *p* (EVI) given the contextually relevant worlds (CONS) is asserted. While this construction *can* be used to add *p* to the common ground,²⁶ it does so unnecessarily indirectly, if EVI and CONS are not relevant. Thinking about it, EVI and CONS are just the truth-conditional equivalents of general principles of cooperativity (EVI) and contextual restriction of the modal base (CONS). Making them explicit only serves a purpose when these are in question.²⁷ That the agent had this evidence available already in a time interval preceding the utterance time is also presupposed by *noch wissen* (qua *noch*) according to our (Rosina – Liefke 2025b) semantics.

Again, *noch wissen*-reports can still be used to convey this information, but I expect them to be used primarily when it is either (i) already common ground that the speaker used to know that or how *p* and what is at issue is whether they still do, or (ii) there is reason to stress the quality of the speaker’s evidence. An instance of (i) could be questions like *weißt du noch?* (‘do you remember [back then]?’) in nostalgic contexts, a case of (ii) could be when the interlocutor believes the negation of *p*, creating a context of justification for the rememberer. (“I swear its true! I saw it with my own eyes! I remember every detail!”)

These kinds of contexts may well arise less often for remembering simple facts (i.e. informationally poor propositions) than for remembering ‘how’ (i.e. informationally rich propositions), where just asserting the message *wie p* directly (as in (10c) for *dass*) is not even a grammatical option. ‘I remember how Grandma was robbed’ does not convey any information about how Grandma was robbed – its purpose is *just* to stress the speaker’s retained evidence (“I *could* tell you how Grandma was robbed”). Therefore, it makes

26 ‘My brother just called’ can introduce the existence of a brother, even if it presupposes it.

27 Our preliminary results from an analysis of speech data collected in psychological experiments (from P8, FOR2812, shared by Thomas Lukaschewski, Nikolai Axmacher, and Albert Newen) support this claim.

no sense to use a *wie/how* sentence outside of a context of doubt in a broad sense at all. This leads to a speculative conclusion that seems paradoxical at first sight: The fact that Blue can be so sure about her (indirect) evidence for *dass p* and no one doubts it may be the reason for the lower than expected ratings of BLUE+DASS.

This reasoning with the semantics of *noch wissen* is at least one possible explanation for the distribution of *sich erinnern*-reports in our experiments. This is another piece of tentative support that maybe *sich erinnern* has “inherited” the presupposition/assertion division of *noch wissen* and its interaction between informational richness and evidence.

The comparison with the forced choice paradigm – where DASS sentences were assigned to BLUE in 64% of the cases – shows that it is not the case that experiential memory is generally the “better” memory, and the competition in (10) must be inactive or overruled by other considerations. While a forced-choice task by definition asks for a relative judgment between the options – in the case at hand RED or BLUE (or PINKIE) – it is unclear what this means for pragmatic competition as it occurs in regular discourse, i.e. which competitions between sentences might get triggered, strengthened, weakened or suppressed by the experimental design.

For example, RED is being more precise by using a *wie* sentence than by using a *dass* sentence, because remembering *wie* is more demanding in terms of evidence, and RED has good evidence for both the informationally poor and the informationally rich proposition. This competition between sentences that RED could utter does not show at all in the rating experiments (“they could have said the more precise thing, but that does not make the less precise one false”), but it is a plausible explanation for the results of the forced choice experiment. Adding artificially the competition between the speakers that is focussed by the format could lead to reasoning along the lines of: “RED could have been more precise, so it is BLUE who uttered the *dass* sentence.” The scale labelling in our rating experiments (‘beschreibt die Situation gar nicht/völlig richtig’, ‘describes the situation not correctly at all / completely correctly’) is aimed at truth-conditional semantics as far as possible (see Zhu – Ahn 2023 for the influence of formulations of the instruction and the scale), so it makes sense that it might block some competitions that show in the forced choice format.²⁸

28 Note, however, that the hypothesized competition in (10) appears active and unaffected by the same instructions and labelling of the scale. An additional effect in the forced choice format could be that some people assume a roughly equal attribution of the sentences to

While these differences in active competitions between formats make intuitive sense as an *a posteriori* explanation, they are extremely hard to falsify. If we already knew the semantics of memory reports, the contrast between our forced choice study and our *noch wissen* rating experiment could provide the necessary evidence. As things actually are, we only know that the two sets of results differ relative to one another, and we have no access to any underlying “true semantics”. In order to make claims like “the forced choice format permits competition in terms of precision”, we have to assume the semantics of *some* expression, which is then competing with others. This semantics can itself not be experimentally supported, if what our experimental methods are sensitive for is what is tested for to begin with.

Of course, competition between vignettes depending on how they are presented and depending on the exact task can be tested. What is unclear is (i) how the competition between vignettes and choices relates to pragmatic competition between sentences; and (ii) which experiment is closer to “the semantic truth out there”, if there is such a thing. If the only methodologically accepted way of empirically supporting a given semantic account is a controlled experiment, this experiment has to have some design by definition. The result of a comparison of methods can therefore only be a relative one. (E.g. “Forced choice experiments give rise to more competition in terms of precision than acceptability rating experiments”, but not “The competition in terms of precision in this forced choice experiment is increased compared to natural conversations.”) Maybe a combination of methods of experiments with corpus analyses can go a part of the way, but essentially, pragmatics is just impossible to control for in a reliable way.

4.2 Experience as evidence

Our semantics predicts that experientiality (i.e. the requirement that one must have directly experienced an event in order to ‘remember how’) is a pragmatic inference from what speakers and hearers consider good evidence. Alternatively, one can write experientiality directly into the semantics (Stephenson 2010). However, experience and evidence are so connected in practice that examples that would show cancellability are situated in very far-fetched worlds. In this section, I put forth an internalist interpretation

the characters, leading to the reasoning that “*some* sentences *must* be said by BLUE”.

of our semantics that predicts that individual beliefs about other people's reliability going into the truth-conditions make it harder to use experimental results as evidence for semantics (in the case of memory predicates and phenomena where similar internalist concerns occur). Furthermore, the concepts of direct experience and of evidence are both philosophically nontrivial, so researchers' philosophical and everyday conceptions play a special role in research on memory reports.

As for the matrix predicate, our experiments detect extremely similar patterns for German *noch wissen*, *sich erinnern* and English *remember*. This is tentative support for the idea that we should aim at a unification of experiential and fact-only remembering and of the semantic core of morpho-syntactically diverse memory predicates, because the more predicates can express both cases in systematic ways, the less plausible it gets that these are diachronically coincidental ambiguities. Our experiments are in line with Informational Richness ((5a), Section 2) as a part of the semantics of reports of experiential remembering, because RED who has evidence for / experience of the informationally rich 'what it was like when Grandma ...' is the best speaker of experiential memory reports. The high ratings for RED+DASS/THAT support the entailment of the *that*-sentence in (5c). Five of our six experiments from Rosina – Liefke (2025a) (two of the ones presented here) are neutral with respect to a possible inference to (5b), a direct experience requirement. The GOLDIE experiment provides tentative support that it might not hold, semantically, and that evidence is a better candidate for what e-markers track: The intermediate ratings for GOLDIE are in principle compatible with Stephenson (2010): Goldie's evidence could be good enough for some participants and not good enough for others, and/or good enough in one scene but not the other. A closer look at the individual responses, however, shows, that – descriptively, for now – a strict ordering of the ratings (RED>GOLDIE>BLUE) is by far the most common amongst participant-scene pairings. I.e. GOLDIE did receive many individual intermediate ratings, not just a mix of high and low ones. This is easier to account for using our semantics than it is for Stephenson (2010).

This argument, however, rests on a specific understanding of 'direct' and 'experience' that a proponent of experientiality as part of the semantics (Stephenson 2010) could reject. For example, they could claim that GOLDIE has direct *enough* experience of Grandma's mishaps, assuming that directness of experience is gradable. Alternatively, the duration/granularity of the event can be questioned: If the event is not over for some participants when GOLDIE enters the story, it makes sense that GOLDIE has direct experience of it. (But then

we would expect Goldie to clearly pattern with Red for one, and with Blue for another group of speakers, which is not what the data show, descriptively.) As a third option for the *advocatus diaboli*, participants could ascribe vicarious memories of the Grandma events to GOLDIE, such that she experienced them in an emotionally direct way. These three options are where it gets very abstract and analytic: Does this hypothetical opponent term something ‘experience’ that I term ‘evidence’? At this point, a reminder of what I understand as evidence is called for.

EVI in our semantics is simply the set of worlds that the agent cannot exclude based on their cumulated evidence.²⁹ My interpretation of ‘evidence’ here is conceptually primitive, so evidence is whatever the speaker (or the hearer, in the case of processing the sentence meaning) takes to be evidence in relation to the topic of *p*. A way to paraphrase this idea of EVI is that it is a variation of the set of doxastic alternatives on which it does not matter what the agent actually believes, but instead what their doxastic state *should* be, given their epistemic standpoint wrt. the topic of *p*, according to the speaker’s norms and beliefs. In order to play the role of evidence, a bit of information has to be accessible to the agent according to the speaker and play some sort of legitimate causal role in reasoning about *p* – again according to the speaker.³⁰ A speaker utters *Ich erinnere mich, wie p* (‘I remember how *p*’) when they think that the informationally rich *p*’ is true *and* that they themselves have retained evidence for *p*’. A hearer accepts this sentence when *they* grant the speaker that they have good evidence for *p*’.

To illustrate this idea of evidence, take the example of a fundamentalist Biblical literalist who takes the fact that the story of Noah is described in the

29 In Rosina – Liefke (2025b), we leave what counts as evidence for a given speaker intentionally subject-dependent and do not say much more about it. Assuming cognitively and philosophically loaded concepts in our semantics is not ideal, of course, if we can avoid it, but I argue that we cannot in this case: Salient alternatives are ‘doxastic alternatives’, ‘justification’ or ‘direct experience’ in the semantics.

30 I do not take these paraphrases to be in tension with my claim that evidence is conceptually primitive. My claim concerns the best modelling choice for a cognitive phenomenon (what “comes first” between evidence, belief, and justification) in the absence – to my best knowledge – of decisive psychological evidence. Someone who has a clearer philosophical conception of, say, belief than of evidence may nonetheless profit from me also defining evidence via belief contrary to the analytic relations in the linguistic model. (This can be compared to the fact that, for explanatory purposes, the plural can be defined as non-singular and singular as non-plural in languages with this simple number system – regardless of which a given theory takes to be morphologically primitive.) I thank an anonymous reviewer for pushing me to clarify this.

Bible to be evidence for the truth of the story. A reviewer asks whether it follows that they can utter *I remember how Noah built his ark and filled it with animals* and suggests that this is not the case, intuitively. This is predicted by our semantic account: What is required for acceptability (due to *how*) is not “evidence for the truth of the story [of Noah]” in this case (which could be captured as a conjunction of some pages of propositions). Rather, the agent must have evidence (in the eyes of the speaker) for the informationally rich proposition *p*, ‘how Noah built his ark and filled it with animals’: For both sentences in this conjunction, the *how* would add so much informational detail that the story alone could not provide it: In the Bible, a lot of information about the event in question is missing: E.g., what was the temperature like? Where was Noah standing relative to the ship? Even if an eyewitness may lack some of these details as well, they will still have much more information about the *how* of the event than the person who has only read a limited number of propositions about it.

However, a related case illustrates the power of the agent-relative nature of evidence as used in our semantics, maybe in the spirit of said reviewer: If a Biblical literalist had something they take to be a true vision of the Noah event – and retained this “evidence in their eyes” for a while –, then our semantics would indeed predict that they can utter *I remember how Noah built his ark and filled it with animals (but I don’t remember how Moses talked to God, because my memory of that vision has faded)*. – All of course relative to their own, idiosyncratic utterance conditions, and, hence, evidence standards. If they meet another Biblical literalist who accepts their visions as evidence, this listener will accept the sentence as true.

In the case of individual remembering, the evidence in question is *internal* evidence, like the scene in my mind (that I have reason to assume is linked to my past in a specific way), or someone else’s ability to activate such a scene in their mind.³¹ This is where the interdisciplinary discussion from the introductory Section 1 enters the picture again. Recall that, cognitively speaking, experiential (‘episodic’) remembering (e.g. of one’s 18th birthday party) involves constructive episodic simulation (Addis et al. 2007) and is phenomenologically similar to quasi-visual imagination. Fact-only (‘semantic’) remembering, on the other hand, (e.g. of the fact that singer Cher’s birthday is May 20) is in line with the metaphor of the human mind as a fact storage and

31 Collective forms of remembering (for example of a nation remembering) could make use of a different, external kind of evidence.

activates different brain areas. In the introduction, I noted that it is surprising that both common German memory predicates – *noch wissen* (‘still know’) and *sich erinnern* (‘REFL remember’) – report both kinds of remembering – especially so given the *know*-morphology of *noch wissen* – and that we even conceptualize them as two subkinds of remembering.

Our (Rosina – Liefke 2025b) semantics – tentatively supported by our (Rosina – Liefke 2025a) experiments – solves this puzzle, but in a way that does not directly map ‘episodic’ remembering to ‘remember how’ and ‘semantic’ remembering to ‘remember that’. Rather, the phenomenal nature of an instance of remembering in the cognitive sense (fact storage versus constructive/episodic) plays a role as a part of the evidence an agent has for an (informationally rich or poor) proposition.

Thinking about what it would have meant for the ‘episodic’/‘semantic’ distinction to be mapped to natural language in the most direct way, there should be expressions relating an agent to the cognitive states of episodic recall and fact recall. There are three problems with this idea: (i) it is unclear whether there is one cognitive state of ‘episodic recall’ to begin with, since the brain activity involved in it is very similar to constructive imagination (Addis et al. 2007), (ii) we would expect there to be predicates ‘remember-how’ and ‘remember-that’ on a direct implementation of that idea, instead of one predicate relating an agent to different complement clauses, and (iii) this does not make sense of the case of cognitively episodic recall licensing also ‘remember that’ (like the high ratings of RED+DASS in our experiment presented in Section 3.1. suggest): If ‘remember that’ was just a direct correlate of ‘semantic’ remembering, the entailment from ‘how’ to ‘that’ would carry on to cognitive states such that one always recalls the fact that the event happened when one episodically constructs a memory. That does not seem to be the case, cognitively (given consciously false memories and the similarity of imagining and episodic remembering, Addis et al. 2007 a.o.). Finally, (iv), speculatively, it also makes more sense that language evolution would track other people’s evidence and reliability rather than what exactly is going on in their brains, since the former is behaviorally important and more accessible.

On my philosophical interpretation of our (Rosina – Liefke 2025b) semantics that relates ‘remembering that’ and ‘remembering how’ to the ‘episodic’/‘semantic’ distinction in a much more indirect way, none of the problems in (i)-(iv) arise: What cognitive scientists and philosophers call ‘episodic memory’ is one real-world case of commonly accepted evidence (combined with some beliefs about why it is not just an imagination) that speakers ascribe to

an agent when they utter ‘*x* remembers how *p*’. Only in the case of ‘episodic memory’ in the cognitive sense do speakers usually find it plausible that someone could have enough evidence for the super informationally rich proposition ‘how things were when *p*’. We have learned this about other people’s cognition, and this made it into natural language. Hence, while there does not seem to be a direct correlate of ‘experiential’ or ‘episodic’ memory in natural language, the fact that many people’s *EVI* is best restricted by direct experience shows that we assume a functioning episodic memory mechanism in other people.

In much of the philosophy of memory, ‘remembering how’ has been understood as necessarily involving a certain phenomenology that is only reached via first-hand experience (e.g. perspectivity and a sense of pastness). Our semantics can be used to show how this philosophical concept of ‘remembering how’ is distinct from the semantics of natural language ‘remembering how’-constructions. The phenomenal character of a given recall does not play a role in whether we utter these sentences or whether we think a speaker says the truth when uttering them. This is supported philosophically by the fact that aphantasic people (who do not recall episodically in terms of the phenomenal experience at all) still utter ‘remember how’-sentences – and no speaker (other than maybe a philosopher making a principled point about a concept) would tell them *No, that’s not true – you are aphantasic*. Philosophically speaking, this means that natural language is closer to folk-psychology than to most debates and concepts in philosophy of mind. A similar point has been made by Dranseika et al (2021) about the phenomenology of observer versus field perspective recall: It does not matter in folk-psychology (I add: or in semantics) how exactly the other person is experiencing something – for the folk (and speakers and listeners), it is the output of it that matters. – Here: Can I trust their evidence?

Abstracting away from our concrete semantic account, the question of whether our experiments support an evidential semantics of memory predicates also highlights a general issue with the semantics of many attitude predicates. What speakers, extensionally speaking, consider a case of knowledge or remembering, varies a lot. Even with our design of Red and Blue as very straightforward cases – involving no contextual doubt, no questioning of their epistemic authority, no memory loss, no contradicting evidence – we get very varied results for the indirect experiencer Blue (see Figures 3–6). The ongoing debate about marginal cases of remembering also bears witness to this inter-personal variation. This puts semanticists in the position to make

a choice between (at least) three options: Either (a) we account for only a small proportion of actual uses of memory reports to be true (so we assume, e.g., that direct experience in a narrow sense is a strict requirement for ‘remembering how’) and claim that speakers are mistaken about ascriptions of remembering all the time, or (b) we take more seriously in these cases the general idea that every speaker, strictly speaking, has their own semantics, and give a variety of idiosyncratic semantics, or (c) we give one general semantic account, but that is so underspecified that it can be filled with many lexical specifications (the investigation of which is then handed over to sociology or experimental philosophy). My interpretation of our (Rosina – Liefke 2025b) semantics is that it is a version of strategy (c). Note that neither the choice between (a), (b) or (c) nor the choice between concrete instantiations of option (c) – e.g. our evidential semantics versus a version of Stephenson (2010) that views direct experience as gradable – is empirically motivated in an obvious way. This is a reminder of the importance of modelling choices in formal semantics. The empirical picture can often be captured by several accounts, the choice between which is often one of methodological and ontological preferences.

Again, as in Section 4.1 with pragmatics in experiments, the relation between evidence and experience on my philosophical interpretation of our semantic account is therefore an *a posteriori* explanation that makes intuitive sense but is extremely hard to test for. One way to go is constructing scenarios and examples in which evidence and experience diverge. Our evidential semantics combined with world knowledge predicts that direct experience is neither necessary nor sufficient in some unusual cases. To show non-necessity, we need a case where there is undeniably no direct experience, but the speaker still grants the agent evidence for *wie p*. Cases going in the direction of (11) are candidates.

- (11) a. Incredible! This mentalist remembers the Goddess creating the first human!
 b. Incredible! This humanoid robot remembers how the titanic sank! She can show you a film she generated from all available information.

As for insufficiency, our semantics (under my interpretation) predicts the challenging of experiential remembering in (12) as acceptable, where there was clearly direct experience and there is a mental scene corresponding to it, but the scene is not accepted as good evidence due to the assumed unreliability of the agent’s general memory system or lack of authority over the topic at hand.

- (12) a. Oh, good to know that that actually happened! I think that I also remember swimming in the Danube at New Year's eve 2020, but it might as well be a hallucination – I was so high that night.
- b. Sure, *we, your parents*, remember how you fell on your birthday cake as a toddler. But you were two years old, you can't possibly remember it. You're just thinking of the video we showed you.

All of these cases remain to be tested, and many of them involve counterfactual human cognition and other scenarios very far from the actual world, which somewhat compromises intuitions. Finally, I also expect (e.g. gender-based and racist) biases about other people's credibility to influence the acceptability of experiential memory reports.

5. Conclusion and open issues

This paper brings together my existing semantic and experimental work co-authored with Kristina Liefke on memory reports involving primarily German *noch wissen, wie* and *noch wissen, dass* ('still know how/that'), with an extension to German *sich erinnern* and English *remember*.

In Rosina – Liefke (2025b), we explain why *noch wissen* – an apparent knowledge predicate – can be used to report experiential remembering of apparent events. We give (pre-theoretic) events expressed by *wie*-complements a propositional type by analyzing them as informationally rich propositions and put forth an evidential notion of knowledge and remembering. As a consequence, we can explain attested differences between *noch wissen, wie* and *noch wissen, dass* via an assumption on the speaker's side that informationally rich propositions require better evidence (e.g. direct experience) in order to reach the threshold for knowledge.

In Rosina – (Liefke 2025a), we report on a series of online experiments which we designed to attest markers of experientiality or – on our semantics – of high evidence requirements, like *wie* ('how') – 'e-markers'. Our rating experiments show that, at least pragmatically speaking, there is a class of e-markers in memory reports that ranges over languages, memory predicates and complement kinds. There *are* morphological elements that render memory reports unacceptable for indirect experiencers.

In one of our rating experiments, we introduced a character who has 'semi-direct' experience of the remembered event. (E.g. she saw a broken leg – does she remember Grandma breaking her leg?) Acceptability judgements

for memory reports with e-markers for this character are in between the ones for the direct and for the indirect experiencer. Unless direct experience is taken to be gradable, this experiment supports our evidence-based semantics and the claim that experientiality is an inference from world knowledge.

The similarities between *noch wissen* and *remember* and *sich erinnern* tentatively support our decomposition of memory predicates into *still+know*. To make this compositional for *remember* and *sich erinnern*, we would have to link this to their morphology. Broad typological data would tell us more about the most common shapes of memory predicates and possible correlations with their use.

For the case of *dass/that* complements in situations where the agent only has been told about the event (so concerning the question whether indirect experiencers remember facts), the results depend heavily on the experimental paradigm. When participants are asked to rate a *dass* memory report uttered by an indirect experiencer, acceptability is much lower than for the direct experiencer uttering the same sentence. When participants are asked to choose who – between direct and indirect experiencer – uttered the *dass* memory report, they choose the indirect experiencer. As an *a posteriori* explanation for this finding, I suggest that the different paradigms suppress different pragmatic effects.

At the inter-disciplinary level, I claim that experiential (‘episodic’) memory as distinct from fact storage in psychology and philosophy is reflected in experiential memory reports in a way that is different from how it initially appeared. *noch wissen, wie* is not a direct natural language correlate of experiential remembering in the cognitive sense. Instead, the philosophical and cognitive distinction only enters the picture at the semantics-pragmatics interface, in the form of speakers’ common belief that ‘experiential memory’ constitutes the best evidence for informationally rich propositions.

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On using experiments on *remember* as evidence in formal semantics

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